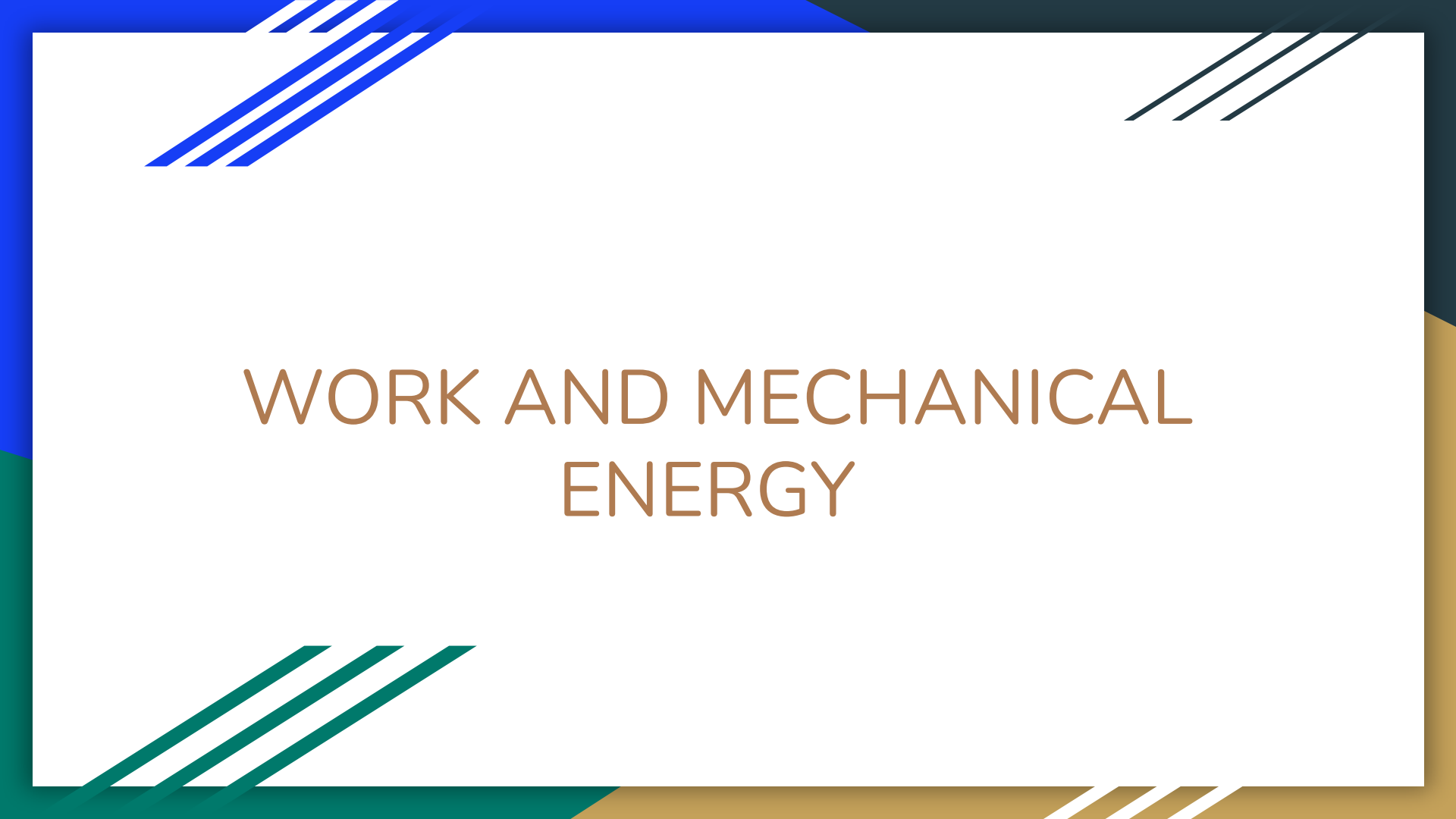




WORK AND MECHANICAL ENERGY

For a system $W_{\text{net}} = \Delta E$ (this equation will work for objects and systems)

This relationship will work whether energy is conserved or not. It all depends on how the system is defined and the external forces that are present.

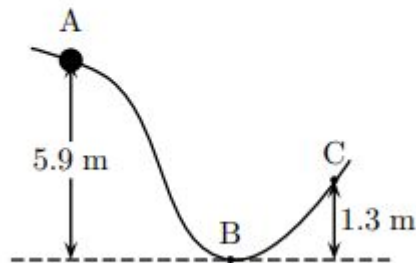
1. Note: If the only forces present in a system are conservative forces, then the mechanical energy of the system is conserved.
2. Note: If external forces act on a system and do no work, then the mechanical energy of the system will be conserved.

A 5 kg object slides down a 30 degree frictionless incline from a height of 5m. Determine its speed at the bottom of the incline.

A 5 kg object slides down a rough 30 degree incline from a height of 5m. Determine its speed at the bottom of the incline. If the coefficient of friction is 0.2.

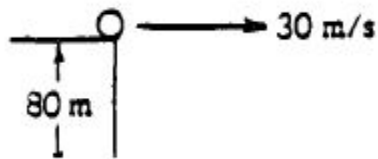
A 5 kg object slides down a rough 30 degree incline from a height of 5m and reaches a speed of 4 m/s. Determine the work done by friction.

1. A 0.4 kg bead slides on a curved wire, starting from rest at point A as shown in the figure. The segment from A to B is frictionless, and the segment from B to C is rough. The point A is at height 5.9 m and the point C is at height 1.3 m with respect to point B.



Find the speed of the bead at B. The acceleration due to gravity is 9.8 m/s^2 .

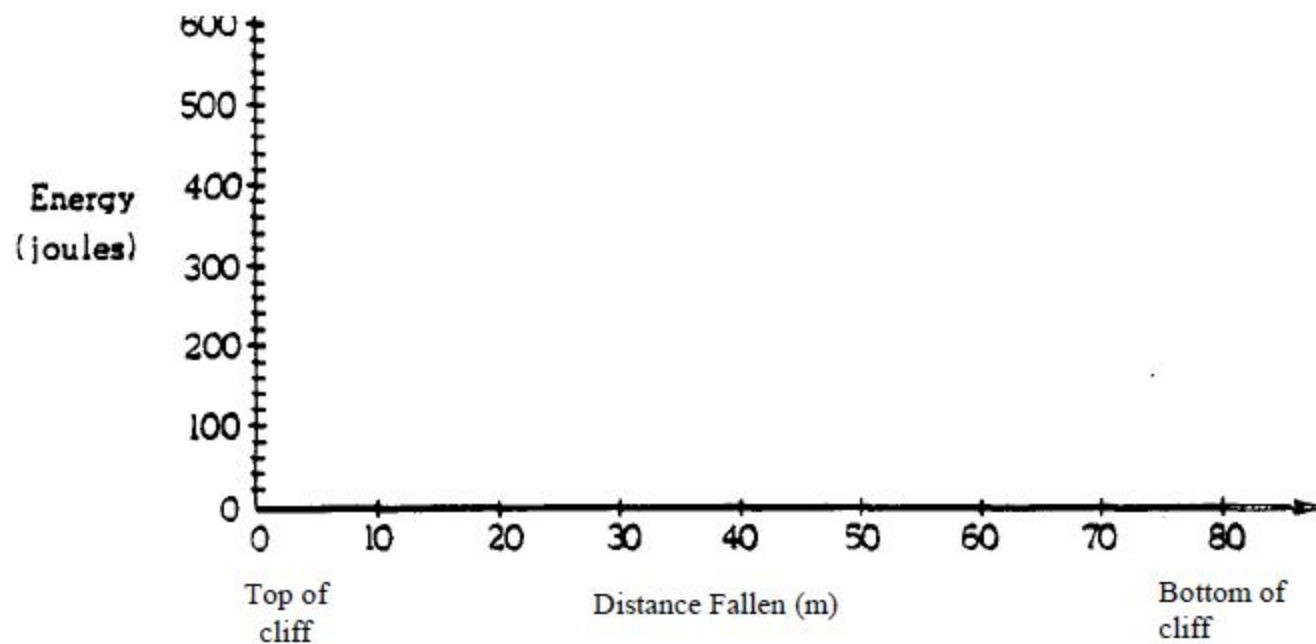
2. If the bead comes to rest at C, find the change in mechanical energy due to friction as it moves from B to C.

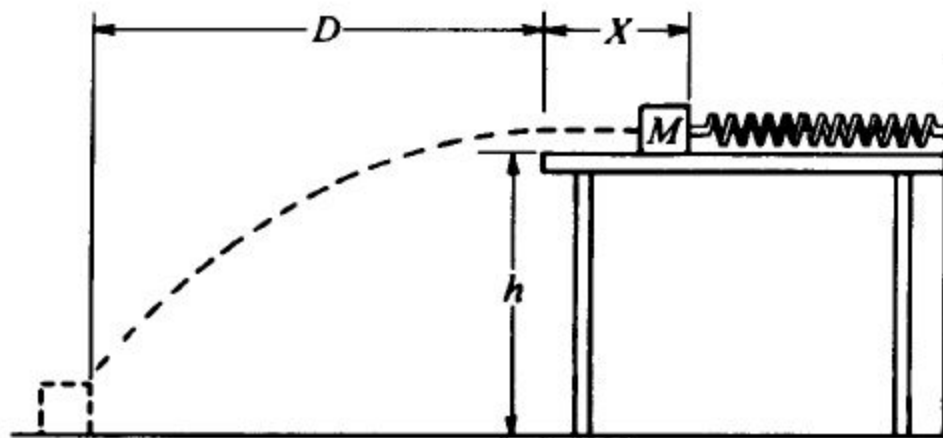


From the top of a cliff 80 meters high, a ball of mass 0.4 kilogram is launched horizontally with a velocity of 30 meters per second at time $t = 0$ as shown above. The potential energy of the ball is zero at the bottom of the cliff. Use $g = 10$ meters per second squared.

- a. Calculate the potential, kinetic, and total energies of the ball at time $t = 0$.

- b. On the axes below, sketch and label graphs of the potential, kinetic, and total energies of the ball as functions of the distance fallen from the top of the cliff

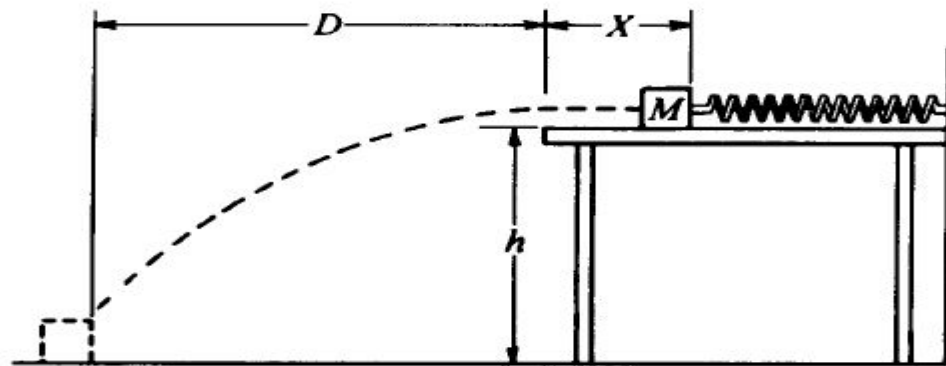




1986B2. One end of a spring is attached to a solid wall while the other end just reaches to the edge of a horizontal, frictionless tabletop, which is a distance h above the floor. A block of mass M is placed against the end of the spring and pushed toward the wall until the spring has been compressed a distance X , as shown above. The block is released, follows the trajectory shown, and strikes the floor a horizontal distance D from the edge of the table. Air resistance is negligible.

Determine expressions for the following quantities in terms of M , X , D , h , and g . Note that these symbols do not include the spring constant.

- The time elapsed from the instant the block leaves the table to the instant it strikes the floor

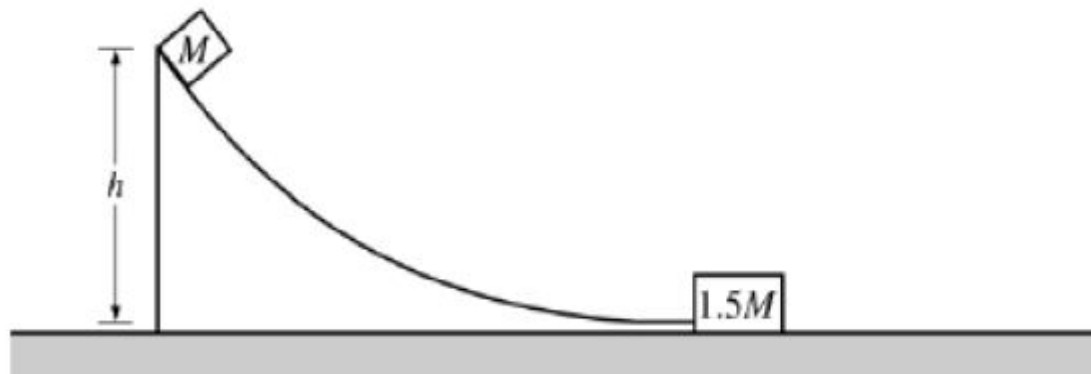


1986B2. One end of a spring is attached to a solid wall while the other end just reaches to the edge of a horizontal, frictionless tabletop, which is a distance h above the floor. A block of mass M is placed against the end of the spring and pushed toward the wall until the spring has been compressed a distance X , as shown above. The block is released, follows the trajectory shown, and strikes the floor a horizontal distance D from the edge of the table. Air resistance is negligible.

Determine expressions for the following quantities in terms of M , X , D , h , and g . Note that these symbols do not include the spring constant.

- b. The horizontal component of the velocity of the block just before it hits the floor

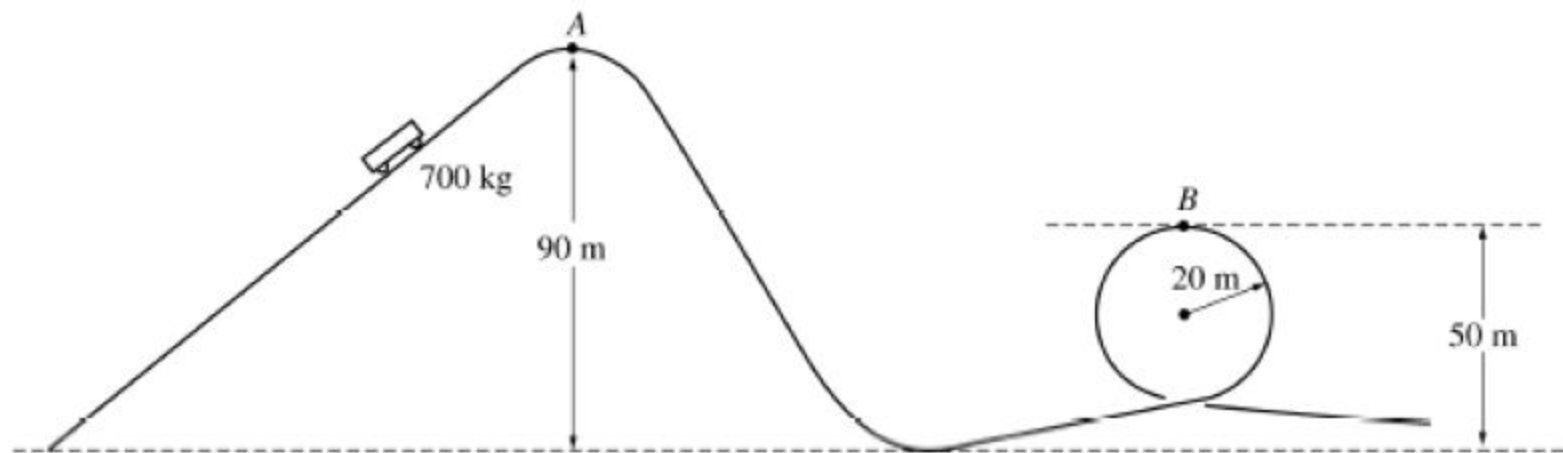
B2006B2



A small block of mass M is released from rest at the top of the curved frictionless ramp shown above. The block slides down the ramp and is moving with a speed $3.5v_o$ when it collides with a larger block of mass $1.5M$ at rest at the bottom of the incline. The larger block moves to the right at a speed $2v_o$ immediately after the collision.

Express your answers to the following questions in terms of the given quantities and fundamental constants.

(a) Determine the height h of the ramp from which the small block was released.

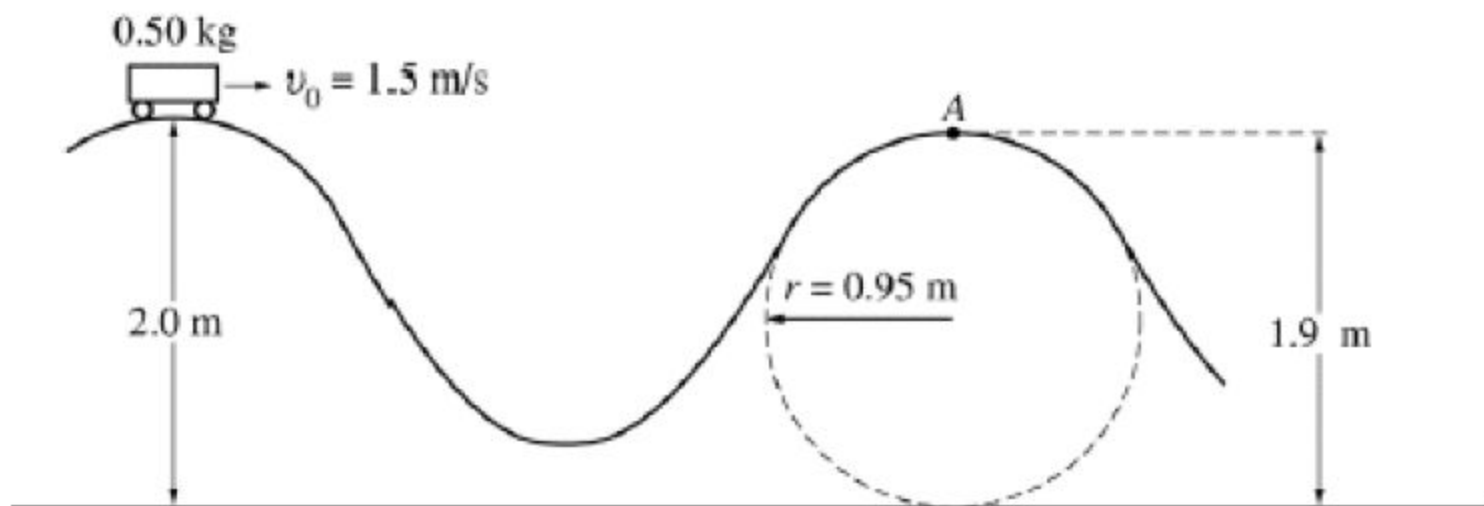


2004B1.

A roller coaster ride at an amusement park lifts a car of mass 700 kg to point A at a height of 90 m above the lowest point on the track, as shown above. The car starts from rest at point A, rolls with negligible friction down the incline and follows the track around a loop of radius 20 m. Point B, the highest point on the loop, is at a height of 50 m above the lowest point on the track.

(a)

- Indicate on the figure the point P at which the maximum speed of the car is attained.
- Calculate the value v_{max} of this maximum speed.



B2004B1.

A designer is working on a new roller coaster, and she begins by making a scale model. On this model, a car of total mass 0.50 kg moves with negligible friction along the track shown in the figure above. The car is given an initial speed $v_0 = 1.5 \text{ m/s}$ at the top of the first hill of height 2.0 m. Point A is located at a height of 1.9 m at the top of the second hill, the upper part of which is a circular arc of radius 0.95 m.

(a) Calculate the speed of the car at point A.

An 5 kg object sliding towards a horizontal spring at 4 m/s collides with the spring and compresses it and comes to a stop. Calculate how much the spring is compressed. The spring constant is 100 N/m.



Compress a Spring 001

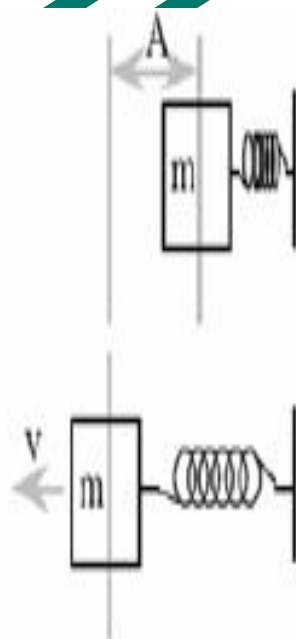
An ore car of mass 40000 kg starts from rest and rolls downhill on tracks from a mine. At the end of the tracks, 10 m lower vertically, is a horizontally situated spring with constant $4 \times 10^5 \text{ N/m}$.

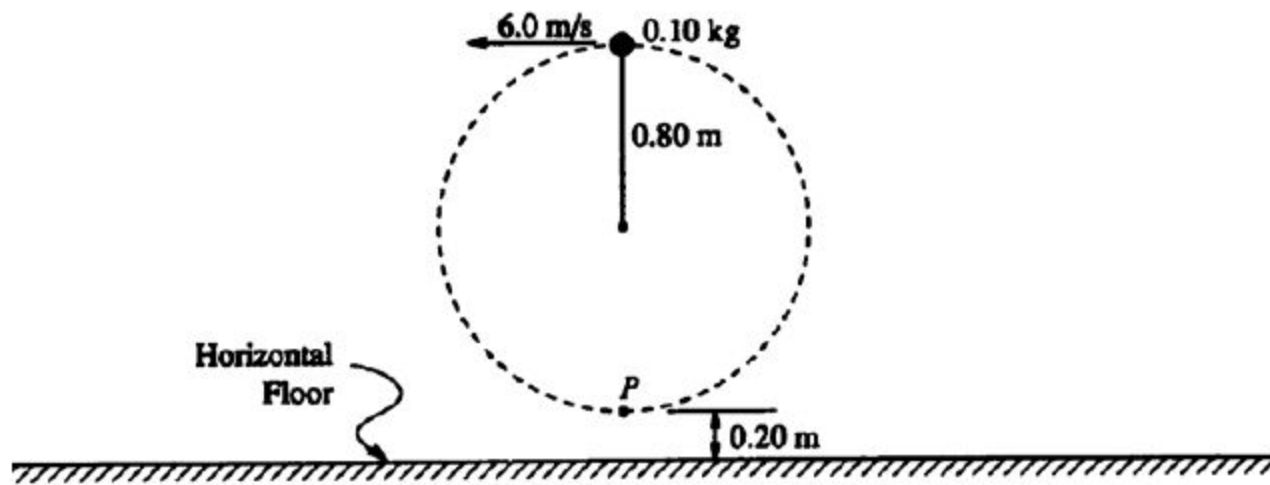
The acceleration of gravity is 9.8 m/s^2 . Ignore friction.

How much is the spring compressed in stopping the ore car?

A mass m attached to a horizontal massless spring with spring constant k , is set into simple harmonic motion. Its maximum displacement from its equilibrium position is A . What is the masses speed as it passes through its equilibrium position?

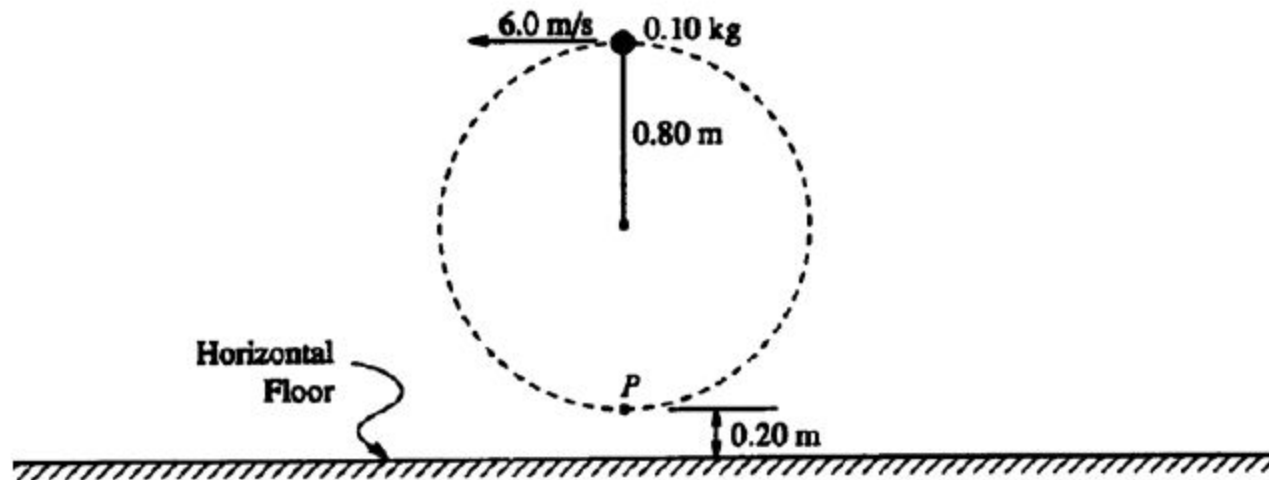
- (A) 0 (B) $A\sqrt{\frac{k}{m}}$ (C) $A\sqrt{\frac{m}{k}}$ (D) $\frac{1}{A}\sqrt{\frac{k}{m}}$





1992B1. A 0.10 -kilogram solid rubber ball is attached to the end of an 0.80 meter length of light thread. The ball is swung in a vertical circle, as shown in the diagram above. Point P , the lowest point of the circle, is 0.20 meter above the floor. The speed of the ball at the top of the circle is 6.0 meters per second, and the total energy of the ball is kept constant.

- a. Determine the total energy of the ball, using the floor as the zero point for gravitational potential energy.



1992B1. A 0.10 -kilogram solid rubber ball is attached to the end of an 0.80 meter length of light thread. The ball is swung in a vertical circle, as shown in the diagram above. Point P , the lowest point of the circle, is 0.20 meter above the floor. The speed of the ball at the top of the circle is 6.0 meters per second, and the total energy of the ball is kept constant.

- Determine the total energy of the ball, using the floor as the zero point for gravitational potential energy.
- Determine the speed of the ball at point P , the lowest point of the circle.

A pendulum consists of a ball of mass m suspended at the end of a massless cord of length L as shown. The pendulum is drawn aside through an angle of 60° with the vertical and released. At the low point of its swing, the speed of the pendulum ball is

- (A) $\sqrt{gL}$ (B) $\sqrt{2gL}$ (C) $\frac{1}{2}gL$ (D) gL

